CARLOS NAZZARIAN

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SUMMARY

Recent Biomedical Engineering graduate with hands-on experience in microcontroller-based systems, electronic instrumentation, PCB design, signal acquisition, and hardware–software integration. Experienced with Arduino, embedded C/C++, Python-based test automation, oscilloscope validation, and laboratory experimentation. Seeking an entry-level role in embedded systems, firmware, electronics, or R&D.

EDUCATION

B.Eng., Biomedical Engineering — Polytechnique Montréal Aug 2021 – May 2025

Relevant Coursework

Circuits (ELE1600), Electromagnetics (PHS1102), Computer Networks (INF3405), Instrumentation & Measurements (GBM2330), Modern Physics for Engineers (GBM3805), Numerical Methods & Signal Processing, Biomedical Imaging, Embedded Instrumentation.

CERTIFICATES

- ***Python Specialization*** – Coursera (University of Michigan), September 2025
- ***Advanced Embedded Systems with Arduino*** – Coursera *(Packt),* September 2025

TECHNICAL SKILLS

- **Embedded Systems & Electronics:** Microcontroller-based measurement and instrumentation systems, Analog and digital signal acquisition, sensor interfacing, PCB schematic design and layout (KiCad), debugging and noise considerations.

- **Instrumentation & Validation:** Oscilloscopes, multimeters and laboratory measurement tools, DAQ systems, calibration and performance verification, Hardware/software troubleshooting and experimental setup design.

- **Programming & Automation:** Python (NumPy, SciPy, Matplotlib) for data anlysis and test automation, C/C++ (fundamentals) and MATLAB for embedded and signal processing tasks, Bash scripting, real-time data acquisiton pipelines.

- **Documentation & Prototyping:** Test plan development, technical documentation and engineering reports, CAD tools (CATIA V5, FreeCAD), microfluidic device fabrication.

RELEVANT ENGINEERING EXPERIENCE

Research Intern – BASyL Laboratory, Polytechnique Montréal (Summer 2024)

PCB Design for Potentiostat

- Designed schematic and PCB layout to reduce noise issues observed in breadboard testing.
- Optimized component placement and routing for improved electrical reliability.
- Performed comparative measurements to validate PCB signal quality.

Microfluidic System Development

- Designed a compact 3D-printed microfluidic chamber for paper-based electrodes.
- Integrated tubing and fastening components; verified fluid flow and sealing.
- Performed voltammetry experiments and analyzed current responses across conditions.

General Responsibilities

- Hardware validation using oscilloscopes and measurement tools.
- Execution of experimental procedures and preparation of engineering documentation.

Laser Speckle Contrast Imaging (LSCI) Project (Winter 2025)

- Developed the optical acquisition setup combining a laser source and camera for speckle imaging.
- Implemented Python scripts for image acquisition and real-time processing.
- Performed tuning of illumination, exposure, and acquisition parameters to improve image quality.
- Executed structured test procedures and produced technical documentation and test reports.

Deep Learning Segmentation in Imaging Project (Winter 2024)

- Prepared microscopy datasets and manual labels for axon segmentation.
- Implemented preprocessing, model training, inference, and morphometric analysis.
- Automated workflows using Python and Bash; evaluated model performance using quantitative metrics

Electrochemical Measurement Instrument — Analog Circuit Design & Testing (Winter 2023)

- Built an analog measurement circuit using op-amps, passive components, and Arduino-generated signals.
- Verified triangular waveform generation and circuit behavior using an oscilloscope.
- Performed voltammetry experiments (dopamine, KCl, ruthenium) and analyzed peak responses.
- Documented system behavior, limitations, and test results.

OTHER PROFESSIONAL EXPERIENCE

Customer Service Attendant – Couche-Tard (May 2022 – Present)

Reliable team member in a fast-paced environment, demonstrating communication, autonomy, and problem-solving.